THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL FULL-YEAR PAPER SET 1

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. Assertion (A): A fair test changes only one factor at a time. Reason (R): Changing several factors together makes it impossible to know which caused the effect. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. Assertion (A): Animal cells burst more easily than plant cells in very dilute surroundings. Reason (R): Animal cells have a rigid cell wall. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. Assertion (A): Antibiotics are not effective against influenza. Reason (R): Influenza is caused by a virus, which has no bacterial structures for antibiotics to act on. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. Assertion (A): An electromagnet loses its magnetism when the current is switched off. Reason (R): Soft iron does not retain magnetism. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. Assertion (A): It is easier to push a box on a floor with rollers than on a rough floor. Reason (R): Rolling friction is less than sliding friction. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. Assertion (A): A knife with a sharp edge cuts easily. Reason (R): A sharp edge exerts large pressure because of its small area. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. Assertion (A): Hot tea cools faster when poured in a wide saucer. Reason (R): A larger surface area increases the rate of evaporation. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. Assertion (A): Water is a compound and not a mixture of hydrogen and oxygen. Reason (R): Hydrogen and oxygen in water are chemically combined in a fixed ratio. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. Assertion (A): A solution of sugar in water does not scatter a beam of light. Reason (R): The particles of the solute are of very small size. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. Assertion (A): The image formed by a plane mirror is virtual. Reason (R): It cannot be obtained on a screen. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

11. Assertion (A): The shadow of a vertical stick is shortest at noon. Reason (R): At noon the Sun is highest in the sky. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

12. Assertion (A): Removing top carnivores from a forest can harm the whole ecosystem. Reason (R): Herbivore numbers increase and overgraze the vegetation. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

13. Assertion (A): The Earth's inner core is solid even though it is extremely hot. Reason (R): The very high pressure at the centre prevents it from melting. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. A student studies the cooling of tea. Temperatures at 0, 2, 4, 6 minutes are 90, 78, 69, 62 degrees C. The fall during the first 2 minutes compared with the last 2 minutes is: 1

(a) equal **(b)** smaller **(c)** zero **(d)** greater

15. A bacterium of length 2 micrometres is viewed with total magnification of 1000x. Its image size is: 1

(a) 20 mm **(b)** 0.2 mm **(c)** 2000 mm **(d)** 2 mm

16. In a school of 500 students, 40 got infected with influenza in a week and 30 more next week. The number infected after two weeks is: 1

(a) 70 **(b)** 30 **(c)** 40 **(d)** 500

17. **Assertion (A):** In a fair test on how oil temperature affects the puffing of a puri, the thickness of the dough must be kept the same. 1
Reason (R): Only one variable should be changed at a time while all other factors are kept constant.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

18. **Assertion (A):** Robert Hooke used the name 'cells' for the box-like compartments he saw in a thin slice of cork. 1
Reason (R): Cork is made of dead cells whose walls remain as empty compartments.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

19. **Assertion (A):** Health means only the absence of disease. 1
Reason (R): Health includes physical, mental and social well-being.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

20. **Assertion (A):** A compass needle placed near a wire deflects when current flows through the wire. 1

Reason (R): An electric current produces a magnetic field around the wire.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 6 × 2 = 12

Answer in about 20–30 words each.

21. A student's pendulum gives 20 swings in 34 s. Find the time period and explain why 20 swings are timed. 2

22. A cell viewed at 500x magnification appears 5 mm long. Find the actual length in micrometres. 2

23. A town of 8000 has 6000 vaccinated persons. Find the percentage unvaccinated and say why it is important to keep it low. 2

24. Explain why a bird sitting on a high-voltage wire is not shocked. 2

25. A body of mass 3 kg has a weight of about 30 N on Earth. What is its mass and weight on the Moon if gravity there is one-sixth? 2

26. Calculate the pressure under a 5 kg bag (weight about 50 N) that rests on a 0.1 m by 0.05 m face. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. Critically evaluate: 'We tested only one type of soil, so all plants grow best in garden soil.' 3

28. A student takes thin peels from an onion and the inner lining of her cheek. List two differences she would see under the microscope. 3

29. A school finds 12 cases of jaundice in a week. List three likely causes and three preventive steps. 3

30. A student connects a coil, a cell and a switch. She places an iron nail inside the coil and the nail picks up pins. List three changes she can make to pick up more pins. 3

31. A box on a floor is pushed with 30 N and moves with a constant velocity. State the friction. If the push is then reduced to 20 N, describe what happens. 3

OR

A cube of side 10 cm weighs 50 N. Find the pressure on a table. If a second identical cube is stacked on it, find the new pressure on the table. 3

32. A student heats 50 g of ice from -10 degrees Celsius. Describe, with temperature values, the four stages of the heating curve till 110 degrees Celsius. 3

OR

A student has a sample of a grey powder. It is attracted by a magnet and gives hydrogen when acid is added. Identify the powder as an element, compound or mixture and explain. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. **Case study.** Meera wants to know whether the amount of water in the dough changes how well a puri puffs. She makes four puris using 20 mL, 30 mL, 40 mL and 50 mL of water for the same quantity of flour, rolls them to equal thickness and fries them in oil at 180 degrees C. She measures the time each takes to puff. 5

(i) Name the independent variable.

(ii) Name the dependent variable.

(iii) Name one variable that Meera has kept constant.

(iv) Write a testable hypothesis for this study and explain why she should repeat the trial.

34. **Case study.** In the school laboratory, Kavya peels a thin layer from an onion, places it on a slide, adds a drop of safranin stain and a cover slip, and observes it under a compound microscope. She sees rectangular cells arranged closely, each with a dark round spot near the edge. 5

(i) Why is the peel kept thin?

(ii) Name the dark round structure.

(iii) Name the outer rigid layer that gives the rectangular shape.

(iv) She then scrapes cells from the inside of her cheek. Name two differences she would see compared with onion cells.

35. In a locality, 40 people suffered from typhoid after a marriage feast. Suggest the likely source, how infection spread, and how a repeat can be prevented. 5

36. Design an experiment to find how the number of turns affects the strength of an electromagnet. State variables, procedure, expected table and conclusion. 5

OR

A student investigates how friction depends on the weight of a block. Design the experiment (variables, procedure, table, expected graph and conclusion). 5

37. A swimming pool 3 m deep is filled with water. A diver goes to a depth of 3 m. Compare the water pressure on the diver at 1 m and 3 m. Explain how this affects the ears and the design of a dam. 5

OR

Design an investigation to find how the temperature affects the rate of diffusion of ink in water. Mention variables, apparatus, observations and conclusion. 5

38. Design an investigation to distinguish between a mixture and a compound using iron and sulphur. Write the procedure, the observations before and after heating and the conclusion. 5

OR

The solubility of a salt is 20 g per 100 g of water at 20 degrees Celsius and 50 g per 100 g of water at 70 degrees Celsius. 150 g of the saturated solution at 70 degrees Celsius is cooled to 20 degrees Celsius. Find the mass of solute at 70, mass of water, and mass of crystals formed. 5

END OF QUESTION PAPER — **Total: 80 Marks**